

KALTAG, AK, USA

WMO: 702006

Lat: 64.327N

Lon: 158.742W

Elev: 55

StdP: 100.66

Time Zone: -9.00 (NAK)

Period: 99-19

WBAN: 26502

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.2	-39.2	-37.2	0.1	-33.4	-36.1	0.1	-33.0	10.8	-15.4	9.4	-13.9	0.4	230	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	25.6	15.9	23.4	15.1	21.6	14.4	17.3	23.5	16.1	21.6	15.1	20.1	2.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.6	10.5	19.0	13.6	9.8	17.8	12.7	9.2	16.9	48.5	23.7	45.1	21.5	42.3	20.2	21.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.6	-43.4	29.1	2.5	1.8	-45.2	30.4	-46.7	31.5	-48.1	32.5	-50.0	33.8		
(5)			-36.7	18.8	1.1	1.1	-37.6	19.6	-38.2	20.3	-38.8	20.9	-39.7	21.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-2.4	-19.7	-14.5	-12.9	-2.8	7.0	13.4	14.4	12.1	7.2	-2.0	-13.2	-18.0
	DBStd	14.18	11.97	9.08	7.18	6.03	4.86	3.13	3.09	3.10	4.03	5.91	8.05	9.59
	HDD10.0	4877	921	686	711	384	116	7	3	16	101	372	695	867
	HDD18.3	7561	1179	919	969	634	353	150	127	195	335	630	945	1125
	CDD10.0	364	0	0	0	0	22	109	139	79	15	0	0	0
	CDD18.3	7	0	0	0	0	0	2	5	1	0	0	0	0
	CDH23.3	183	0	0	0	0	14	57	92	16	4	0	0	0
	CDH26.7	24	0	0	0	0	1	6	15	2	1	0	0	0
Wind	WSAvg	2.2	2.3	2.2	2.6	2.4	2.2	2.2	2.0	1.7	1.8	1.9	2.2	2.2
Precipitation	PrecAvg	525	36	34	27	21	26	40	63	84	64	48	42	38
	PrecMax	654	75	92	74	48	48	67	119	148	110	136	112	90
	PrecMin	373	10	9	2	2	8	17	22	35	29	14	4	11
	PrecStd	82	18	19	17	12	10	14	23	24	25	26	25	19
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	6.0	3.3	12.7	25.5	27.4	28.8	26.0	21.9	10.2	2.8	2.3
		MCWB	1.5	1.7	0.5	6.1	13.8	16.2	18.9	17.8	13.6	6.5	1.5	0.7
	2%	DB	1.0	1.5	1.3	9.7	21.5	24.9	25.8	22.2	17.4	8.3	1.2	-0.1
		MCWB	0.3	-0.1	-0.5	4.4	11.9	14.8	16.8	16.1	11.7	5.8	0.3	-0.8
	5%	DB	-1.3	-0.4	-0.2	7.8	18.5	22.9	23.7	19.6	15.1	6.9	-0.2	-2.6
		MCWB	-2.4	-1.7	-1.9	3.4	10.2	14.1	15.9	14.6	10.6	4.9	-0.9	-4.0
	10%	DB	-4.0	-2.5	-2.2	5.8	16.1	21.2	21.7	17.8	13.2	5.2	-2.4	-5.3
		MCWB	-5.0	-3.7	-3.6	2.2	9.3	13.5	15.4	13.7	10.0	3.8	-3.2	-6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.4	1.4	6.4	14.1	17.8	19.3	18.5	14.2	8.0	1.7	1.3
		MCDB	3.1	4.9	2.7	12.0	24.6	25.3	27.1	24.0	20.2	9.3	2.3	1.9
	2%	WB	0.4	0.3	0.0	4.7	12.4	16.2	17.7	16.7	12.3	6.4	0.4	-0.8
		MCDB	1.2	1.4	1.1	9.0	20.3	22.7	24.5	21.1	16.0	7.8	1.0	-0.1
	5%	WB	-2.0	-1.5	-1.9	3.6	11.1	15.1	16.5	15.4	11.3	5.1	-1.0	-3.5
		MCDB	-1.1	-0.4	-0.5	7.3	17.1	21.1	22.5	18.8	14.1	6.6	-0.3	-2.5
	10%	WB	-4.3	-3.6	-3.6	2.6	9.8	14.0	15.5	14.3	10.4	3.8	-3.2	-6.0
		MCDB	-3.3	-2.3	-2.0	5.5	15.1	19.5	20.9	17.3	12.9	5.1	-2.3	-5.0
Mean Daily Temperature Range	5% DB	MDBR	8.0	9.8	11.9	10.9	11.7	12.1	10.1	9.5	8.9	6.6	7.5	7.8
		MCDBR	7.9	8.9	10.7	12.3	16.9	16.1	15.3	12.9	12.8	7.3	5.2	6.9
		MCWBR	6.9	7.6	8.8	7.6	8.6	7.7	7.7	7.4	7.9	5.3	4.8	6.2
	5% WB	MCDBR	7.3	8.7	9.2	10.8	14.8	13.9	13.8	11.0	10.4	6.2	5.2	6.3
		MCWBR	6.4	7.7	7.7	6.9	7.7	7.2	7.2	6.7	6.7	4.8	4.9	5.8
Clear-Sky Solar Irradiance	taub	0.184	0.256	0.263	0.323	0.344	0.361	0.365	0.364	0.309	0.280	0.184	0.182	
	taud	1.942	2.170	2.259	2.164	2.364	2.378	2.407	2.426	2.551	2.373	1.948	1.916	
	Ebn at Noon	387	661	837	845	865	854	840	804	792	639	367	140	
	Edh at Noon	26	58	81	114	104	105	100	89	64	48	24	10	
All-Sky Solar Radiation	RadAvg	0.18	0.80	2.58	4.47	5.27	5.42	4.50	3.43	2.17	0.90	0.26	0.06	
	RadStd	0.02	0.10	0.23	0.28	0.56	0.49	0.31	0.40	0.26	0.14	0.04	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page